Clínica Universidad de Navarra

TBL 6.4

Glucose metabolism and diabetes

Dr. Nerea Varo
Clinical Biochemistry Laboratory
Clínica Universidad de Navarra

Contents

- Plasma glucose regulation
- Tests for DM diagnosis
- Tests for the diagnosis of GDM
- Tests for long term follow-up

Glucose homeostasis

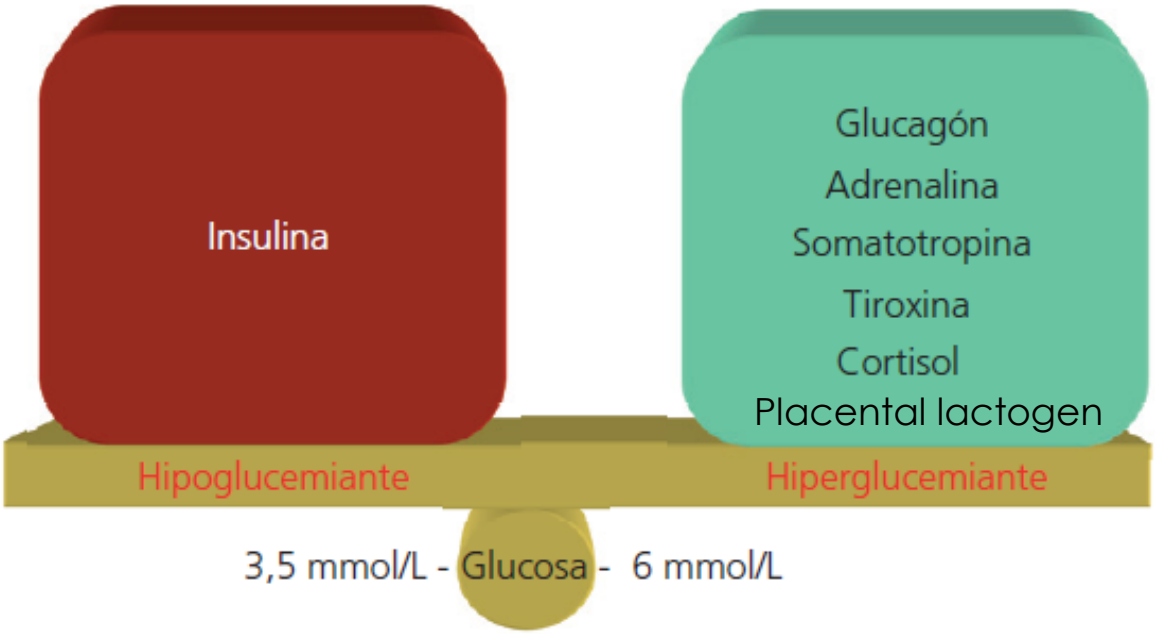
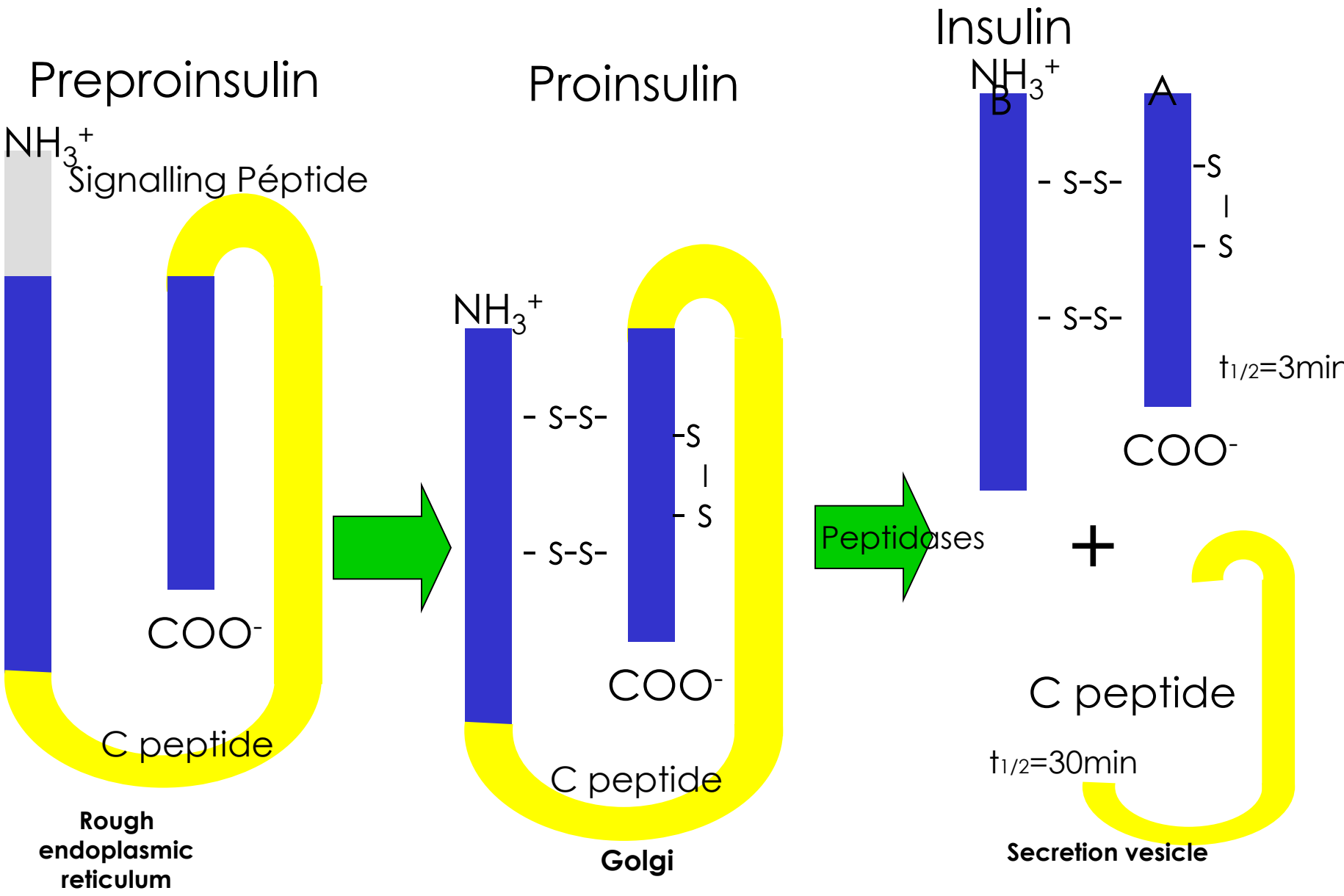


Figura 14-3 Regulación hormonal de la homeostasia de la glucosa.

Plasma reference interval: 80-110 mg/dL (4.4-6 mmol/L)

Insulin synthesis



Diabetes Mellitus (DM)

- Definition
- Causes
- Consecuences
- Symptoms
- Types of DM: ADA/WHO

TABLE 31.1 Table 1 versus type 2 diabetes mellitus

Main features	Type 1	Type 2
Epidemiology		
Frequency in Northern Europe	0.7–0.8%	6–7%
Predominance	Northern European	Worldwide
	Caucasians	Lowest in rural areas of developing countries
Clinical characteristics		
Age at diagnosis	<30 years (peak 9–14 y)	>40 years (but emerging in younger people)
Weight	Low/normal	Increased
Onset	Rapid	Slow
Ketosis	Common	Under stress
Endogenous insulin	Low/absent	Present but insufficient
HLA associations	Yes	No
Islet cell antibodies	Yes	No
Pathophysiology		
Aetiology	Autoimmune destruction of pancreatic islet cells	Insulin resistance that eventually overwhelms insulin secretory capacity
Genetic associations	Polygenic	Strong
Environmental factors	Viruses and toxins implicated	Obesity, physical inactivity

Criteria for the diagnosis of diabetes

Table 2.2—Criteria for the diagnosis of diabetes

FPG ≥126 mg/dL (7.0 mmol/L). Fasting is defined as no caloric intake for at least 8 h.*
OR
2-h PG ≥200 mg/dL (11.1 mmol/L) during OGTT. The test should be performed as described by WHO, using a glucose load containing the equivalent of 75 g anhydrous glucose dissolved in water.*
OR
A1C ≥6.5% (48 mmol/mol). The test should be performed in a laboratory using a method that is NGSP certified and standardized to the DCCT assay.*
OR
In a patient with classic symptoms of hyperglycemia or hyperglycemic crisis, a random plasma glucose ≥200 mg/dL (11.1 mmol/L).

DCCT, Diabetes Control and Complications Trial; FPG, fasting plasma glucose; OGTT, oral glucose tolerance test; WHO, World Health Organization; 2-h PG, 2-h plasma glucose. *In the absence of unequivocal hyperglycemia, diagnosis requires two abnormal test results from the same sample or in two separate test samples.

Diabetes Care 2024

Criteria for diagnosis of prediabetes

Table 2.5—Criteria defining prediabetes*

FPG 100 mg/dL (5.6 mmol/L) to 125 mg/dL (6.9 mmol/L) (IFG)
OR
2-h PG during 75-g OGTT 140 mg/dL (7.8 mmol/L) to 199 mg/dL (11.0 mmol/L) (IGT)
OR
A1C 5.7–6.4% (39–47 mmol/mol)

FPG, fasting plasma glucose; IFG, impaired fasting glucose; IGT, impaired glucose tolerance; OGTT, oral glucose tolerance test; 2-h PG, 2-h plasma glucose. *For all three tests, risk is continuous, extending below the lower limit of the range and becoming disproportionately greater at the higher end of the range.

Diabetes Care 2022

American Diabetes Association

	Normal	Prediab	Diabetes
Ayunas	**<110 WHO <100 ADA**	**110-126 (IFG)**	**>126**
2h*	**<140**	**140-200 (IGT)**	**>200**

* After 75 g of glucose

HbA1c= 5.7-6.4%

Tests for the diagnosis of DM and prediabetes

1. Blood glucose (fasting / random)
2. OGTT (75 g)
3. Glycohemoglobin A1c

1. Blood glucose

Obtention: fasting o random

Serum or plasma

NaF

Separating gel

Spectrofotometric method

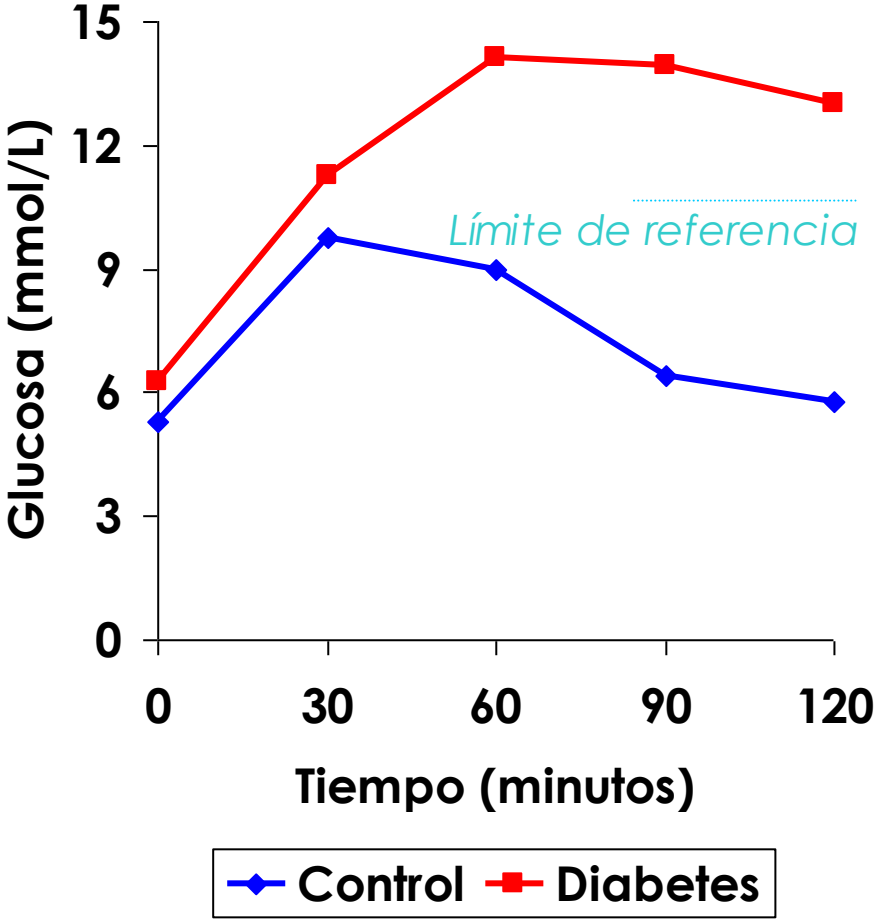
2. Oral Glucose Tolerance Test

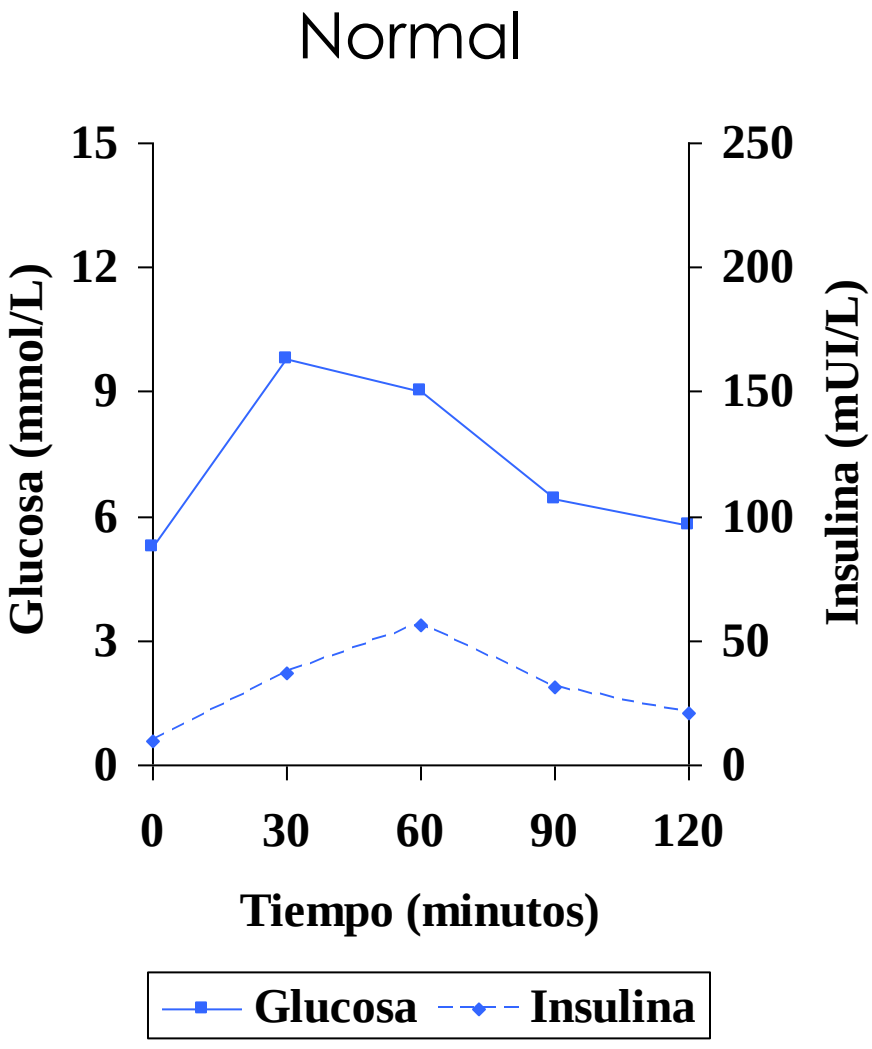
Patient preparation: prior to and during the test

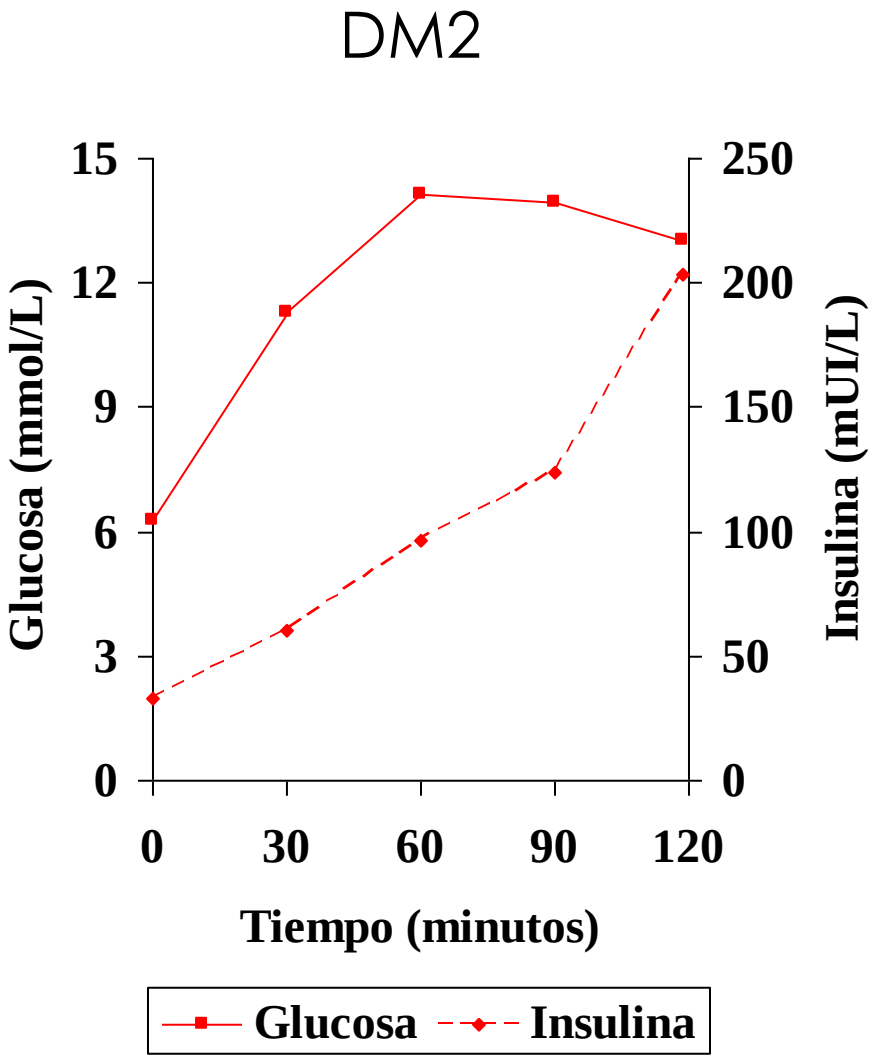
- baseline
  stimulus: 75 g oral glucose
- at 30', 60', 90' y ***120'***

Very often insulin is measured at the same time

OGTT (75 g)


15
12
9
6
3
0
Glucosa (mmol/L)
Límite de referencia
0
30
60
90
120
Tiempo (minutos)
Control
Diabetes


Normal
15
12
9
6
3
0
250
200
150
100
50
0
Glucosa (mmol/L)
Insulina (mUI/L)
0
30
60
90
120
Tiempo (minutos)
Glucosa
Insulina

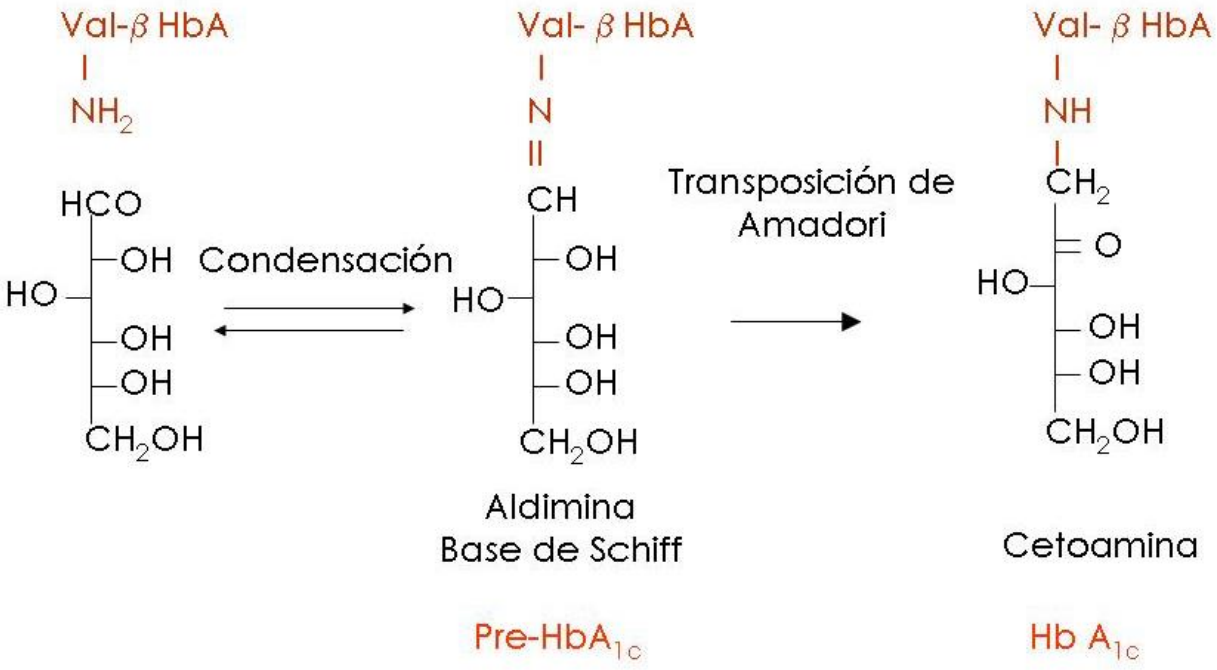

DM2
15
12
9
6
3
0
250
200
150
100
50
0
Glucosa (mmol/L)
Insulina (mUI/L)
0
30
60
90
120
Tiempo (minutos)
Glucosa
Insulina

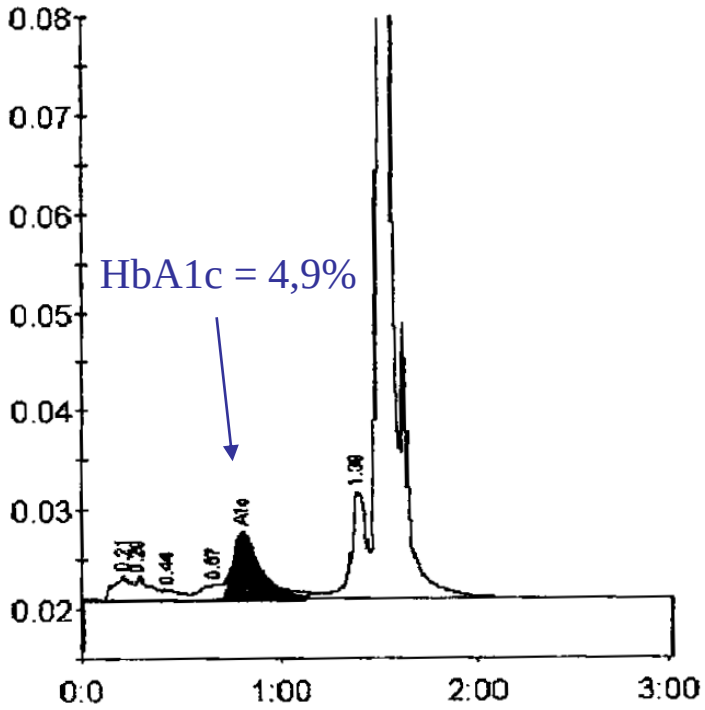
3. *Glycohemoglobin A1c (glycated hemoglobin)*

Definition:

Sample: EDTA

Method: HPLC or immunoassay





Good glycemic control

3. Glycohemoglobin A1c

Advantages

Mean plasma glucose (2-3 months)

No fasting required

Limitations

Shorter eritrocytes halflife

Hemoglobin variants

Gestacional Diabetes Mellitus (DMG)

- Phisiological increase of insulin resistance
- Increases of placental lactogen
- 2-4 % of pregnant women
- Risk factors: family history of diabetes, obesity, age, glucosuria, alterations in previous pregnancies
- After birth: normalization or develop DM

Tests for screening and diagnosis of GDM

One step strategy → • OGTT (75 g) Specific cut-offs

Two steps strategy

- screening → • OGTT (50 g) No fasting
- confirmation → • OGTT (100 g)

Tests for long term follow-up

1. Capillary blood glucose
2. Glycohemoglobine A1c
3. Insulin
4. C peptide
5. Glucagon test

Glycemic control

6. Albumin in urine

Kidney damage

7. Lipid profile
8. Inflammatory and prothrombotic markers

Cardiovascular risk/damage

1. Capillary blood glucose. Self monitoring



POCT

Good correlation with plasma glucose

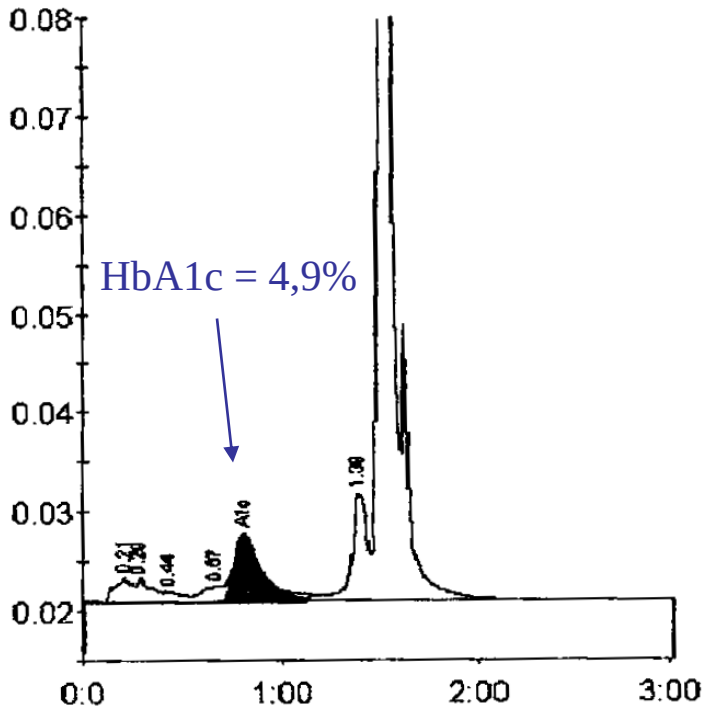
Follow-up at home

https://www.youtube.com/watch?v=Q6UWI4NQoU0

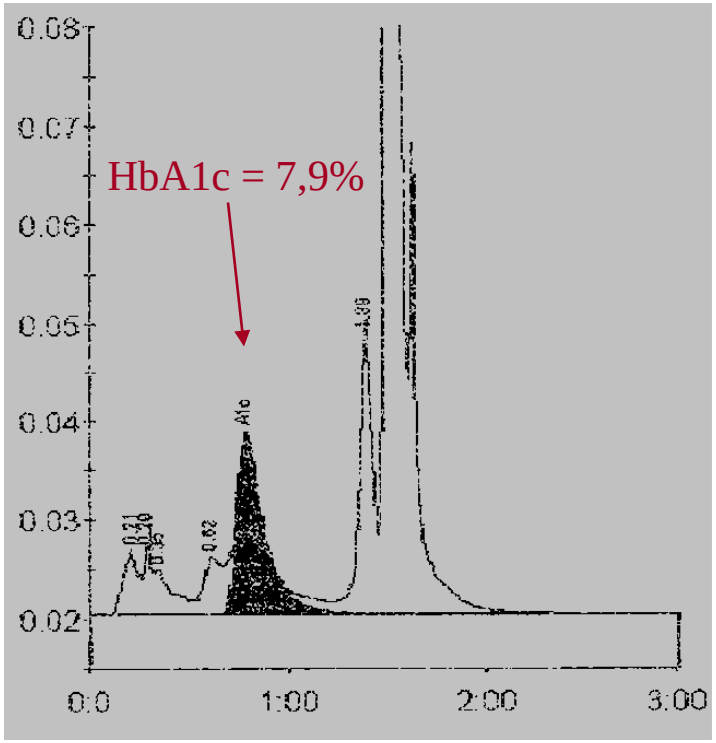
2. *Glycohemoglobin A1c*

7%

Good glycemic control | Poor glycemic control



Tiempo de retención (min)



Tiempo de retención (min)

Good glycemic control

Poor glycemic control

3. *Insulin vs péptido C*

Clinical usefullness: in type 1 DM on treatment with insulin

Half life 30’

5. Glucagon-glucose test

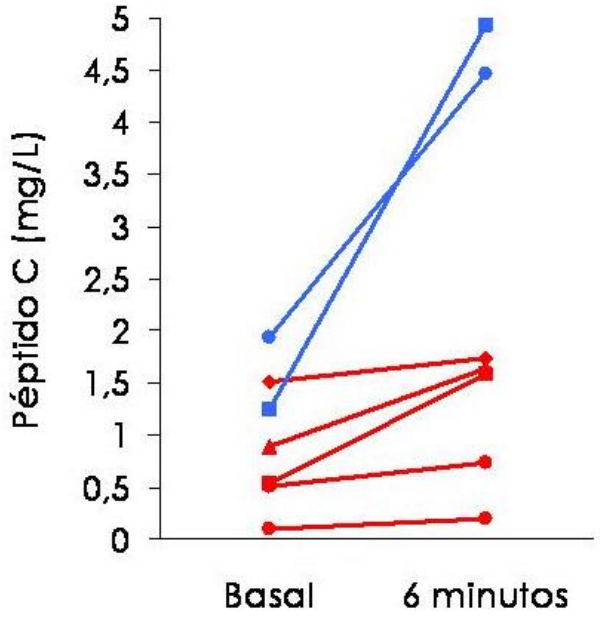
- Pancreatic insulin reserve

- Clinical usefullness : type 2 DM treated with oral hypoglucemic agents

- Samples
  - Baseline

  Estimulus: 1 mg glucagón iv
  - At 6'

- Parameters: glucose and C peptide



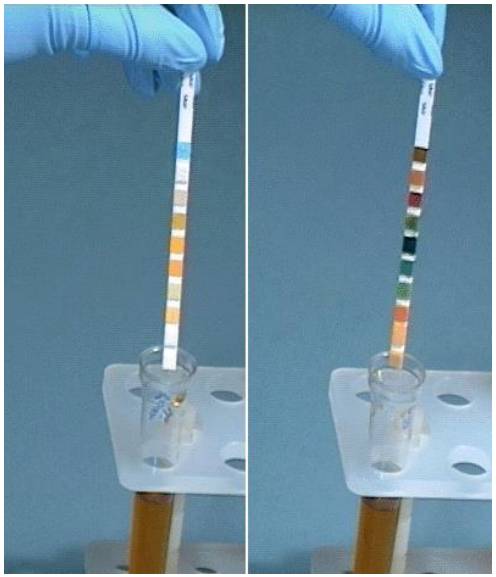
6. *Albumin in urine*

Early kidney disfunction

Urinary test strip

Sample: 24h urine/ spot urine





EVALUATION OF CARDIOVASCULAR RISK

7. Lipid profile

TBL 6.3

8. Inflammatory and prothrombotic markers

C Reactive protein: TBL 6.4

SUMMARY

Testing in diabetes

1. Blood glucose is regulated by insulin and other counterregulatory hormones.
2. Diabetes mellitus is characterised by hyperglycaemia and an absolute or relative lack of insulin.
3. The diagnosis of diabetes mellitus is made on the basis of plasma glucose concentrations or glycated haemoglobin comparing the results with clinical decision limits.
4. Several laboratory tests are employed for early detection of late complications.
5. Self glucose monitoring helps in the diabetes control.